Project Theme

Data-Driven Stochastic Modeling and Simulation of Real-World Systems

Group Size

2-3 students per group

Duration: 6 Weeks

1. Project Objectives

Each group will select a real-world problem involving uncertainty or randomness and develop a mathematical and computational stochastic model to study it.

The project should demonstrate the complete pipeline:

Real-world problem → Data → Probability model → Stochastic process → Mathematical analysis → Simulation → Validation → Interpretation

Students are expected to:

Identify a real-world phenomenon that can be modeled using stochastic processes.
Formulate the problem mathematically.
Select an appropriate stochastic model.
Derive important theoretical properties of the model.
Implement the model using Python/R/MATLAB.
Use real data.
Compare theoretical and empirical results.
Interpret the results in the context of the application.
Present limitations and possible extensions.

2.

Domain	Possible Project	Stochastic Models
Weather & Climate	Weather-state prediction using Markov chains	Markov Chain, Time Series
Air Pollution	Modeling pollution exceedance events	Poisson Process, Markov Chain
Traffic	Vehicle arrival and congestion modeling	Poisson Process, Markov Chain
Hospital	Patient arrivals and waiting times	Poisson/Queueing Models
Bank/Service System	Customer arrival and service analysis	Poisson Process, Queueing
Stock Market	Random-walk model of price movement	Random Walk, Time Series
Sports Analytics	Modeling winning/losing states	Markov Chain
Disease Dynamics	Disease-state transitions	Markov Chain
Social Media	Arrival of posts/messages	Poisson Process
Cybersecurity	Modeling cyberattack/event arrivals	Poisson Process
Reliability	Component failure and replacement	Poisson Process/Markov Models
Renewable Energy	Wind/solar state modeling	Markov Chain, Time Series
Water Resources	Rainfall/drought-state modeling	Markov Chain, Time Series
Gambling	Gambler's ruin	Random Walk
Population Dynamics	Birth/death-type stochastic modeling	Markov Process
Environmental Monitoring	Extreme pollution-event modeling	Poisson Process/Markov Chain

3. Required Project Components

Every group should submit the following.

A. Problem Definition

Clearly explain:

What is the real-world problem?
Why is randomness involved?
What question are you trying to answer?

B. Mathematical Formulation

Students must identify:

Random variables
Stochastic process
State space
Time parameter
Model assumptions
Relevant probability distributions

C. Theoretical Analysis

Students should derive or explain relevant mathematical results.

For example:

Markov Chain

Transition matrix
n-step transition probabilities
Stationary distribution etc

Poisson Process

Inter-arrival distribution
Event-count distribution
Expected number of events etc

4. Computational Component

Every project must contain a substantial computational component.

Students should:

Write their own Python/R code
Simulate the stochastic process
Generate multiple sample paths
Estimate model parameters
Compare simulation with theoretical results
Produce meaningful visualizations

Examples:

Poisson Process

Event-count plot
Inter-arrival histogram
Empirical vs theoretical distribution etc

Markov Chain

State-transition diagram
Transition matrix heatmap
State probabilities vs time
Long-run distribution etc

5. Real Data Requirement

Each group should obtain data from a publicly available source.

Possible sources:

Government datasets
WHO
World Bank
Meteorological data
Stock-market data
Environmental monitoring data
Traffic datasets
Hospital/public-health datasets etc

6. Parameter Estimation & Model Validation

This is something you should do research in. How to estimate parameters of a model? How to validate your model?

You should quantitatively investigate the agreement.

7. Expected Final Deliverable

Report: 10–12 pages

Suggested structure:

Title
Abstract
Introduction
Problem Statement
Literature Review
Data Description
Mathematical Background
Stochastic Model
Methodology
Computational Implementation
Simulation Results
Real Data Results
Model Validation
Discussion
Limitations
Conclusion
References
Appendix – Code

8. Presentation

15-minute group presentation

Recommended structure:

1 min – Introduction & Problem
1 min – Data
2 min – Mathematical model
2 min – Methodology/code
3 min – Results
1 min – Interpretation
1 min – Conclusions & limitations
1 min - Individual Contribution
4 min - Q&A

Students should demonstrate their code and visualizations, consistent with the existing requirement for a 15-minute presentation with visuals and code output.